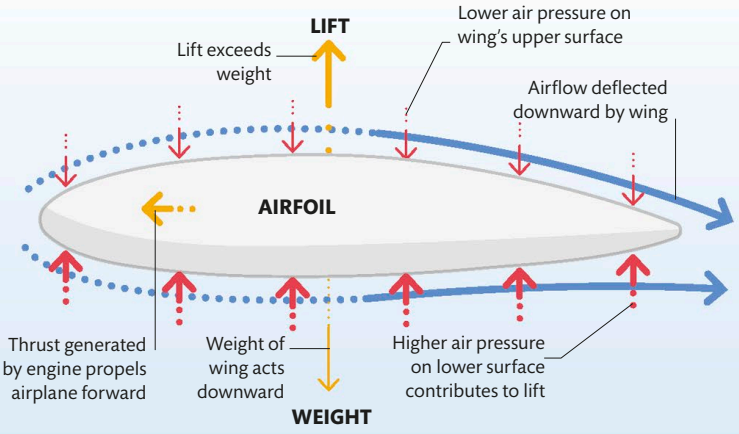
Airplanes

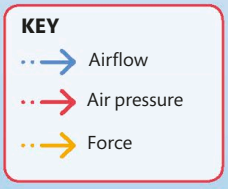
Airplanes come in a wide variety of shapes and sizes, but they all fly according to the same principles. Power generated by an engine or propeller thrusts the plane forward, while wings generate lift.

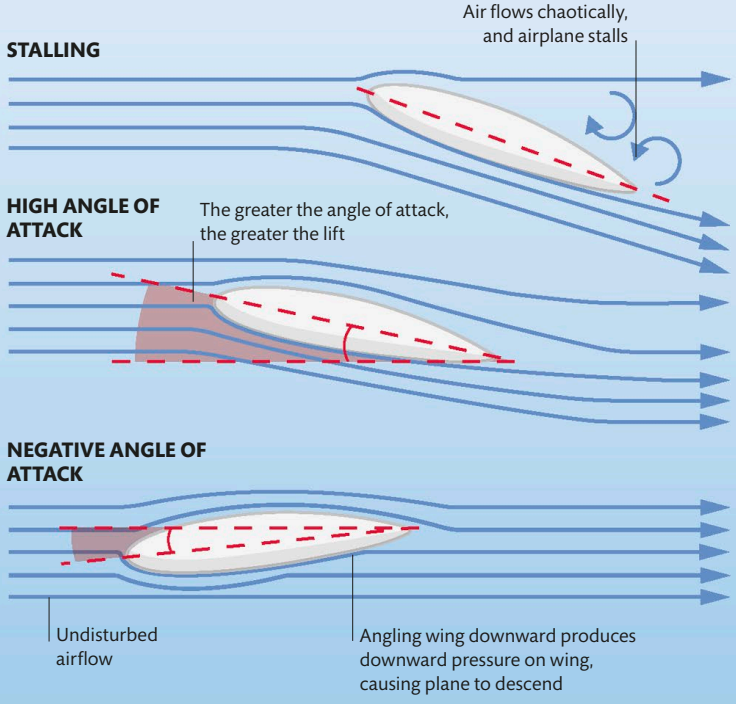


How an airplane flies

When an airplane is propelled forward by its engines (see pp.60–61), its wings slice through the air. A wing's shape, called an airfoil, deflects the air downward. When the wing pushes air down, the air acts in accordance with Isaac Newton's third law of motion by pushing back and producing an upward reaction force known as lift. The air pressure above the wing falls, and the pressure below it rises, contributing to lift generated.

Angle of attack
The angle between a wing and the oncoming air is called the angle of attack. By increasing this angle, more lift is created. If the angle is too big, the airflow separates from the wing, which loses lift or stalls.



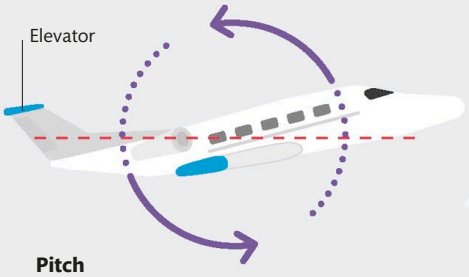


THE AIRBUS A380, THE WORLD'S LARGEST AIRLINER, HAS 4 MILLION PARTS

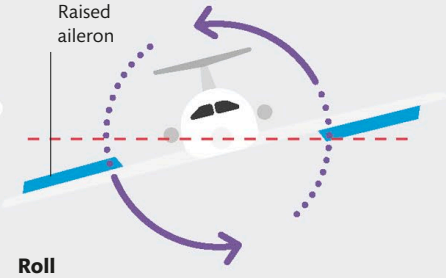


Controlling an airplane

An airplane is steered by moving panels in the wings and tail called control surfaces. There are three types—elevators, ailerons, and a rudder. When the pilot moves the flight controls, the control surfaces move out into the air flowing past the plane, which rotates the plane in three ways—pitch, roll, and yaw.



Pitch
Elevators in the tail's horizontal stabilizers tilt up and down. Tilting them up pushes the tail down, and the plane climbs. Tilting them down makes the plane dive.



Roll
The aileron in one wing is raised, while the aileron in the other wing is lowered. This makes the first wing fall and the second rise, making the plane roll.